

BUSINESS REPORT

CAPSTONE PROJECT (BANKING)

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G4 PGDSBA.O.May 23.A

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1. Brief Introduction about the problem statement and the need of solving it:

a) Problem Statement: The business problem is based on data of customers of a Credit Card company. The Problem Statement is to predict the probability of default of a customer based on the variables provided. The variables are based on age, purchase and delinquency information of the customer.

b) Need of the study/project: The need of the Project is to understand the riskiness of the customers and understand how much credit is at stake in case of customer defaults. Businesswise, this factor is extremely important for a lending company for the purpose of its profitability as NPAs greatly affect the bottom line. Further, provisioning for NPAs and bad loans is required to be kept in company's account books.

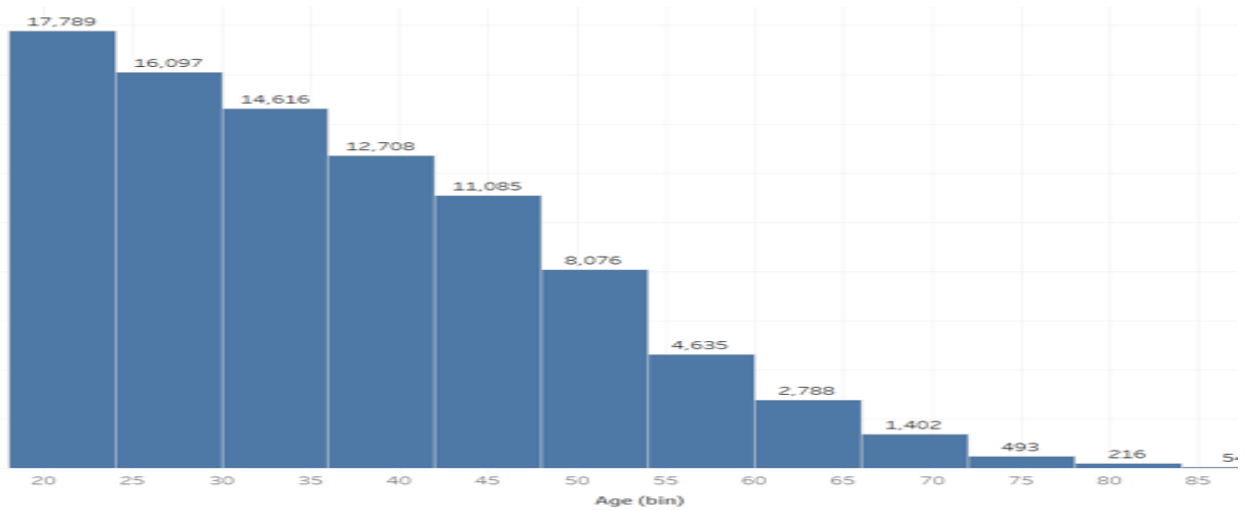
Dataset : PD_modelling_dataset ; The dataset is renamed as Capstone_Dataset_Banking for ease.

2. EDA and Business Implication:

Data Report:

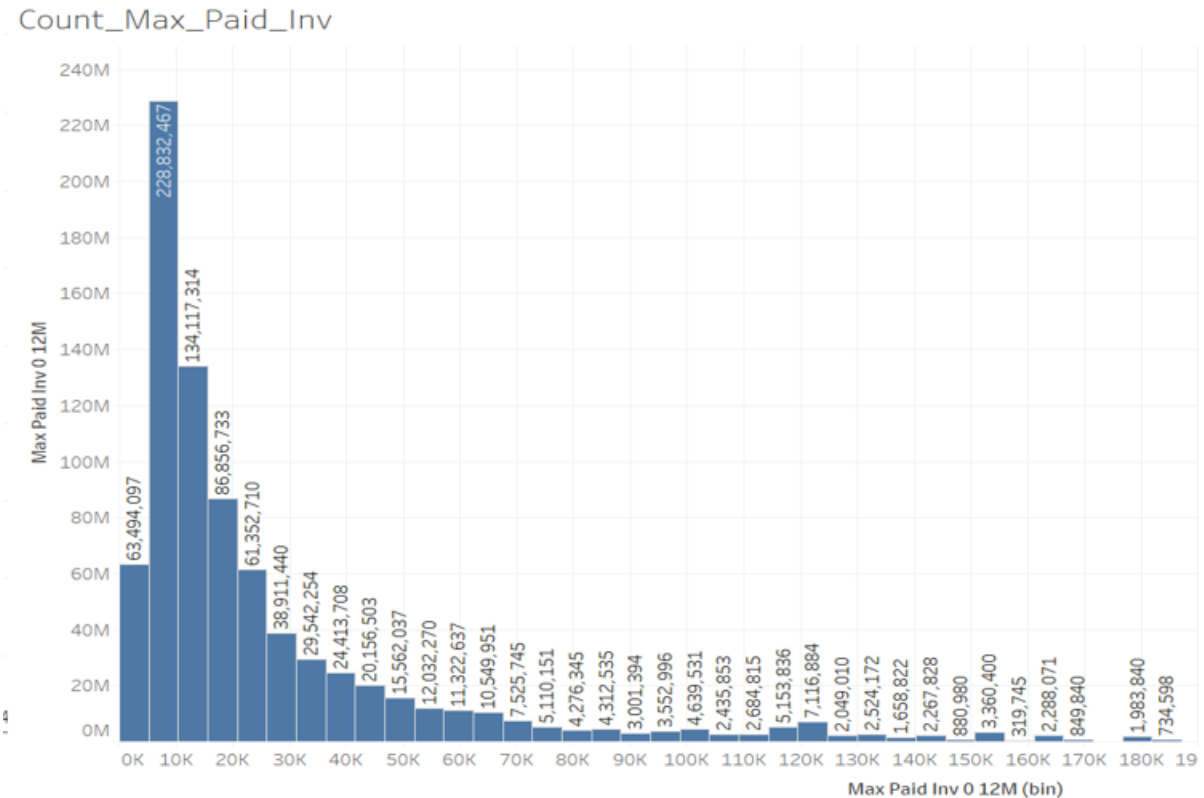
- a) **Data Collection:** The data was collected over a period of 24 months for 99979 customers with respect to User ID and Default status. The data primarily consists of age, account status, customer spendings and default status of various customers.
- b) **Visual inspection of data:** The data has 99979 rows (data points) and 36 Columns (variables).
- c) **Understanding of attributes:** Out of 36 variables/attributes, 31 are float type and 5 are object type variables. 'default' is the target variable of the data. No variables have been renamed so as to maintain original status for ease of understanding.
- d) **Univariate Analysis:** Charts were plotted for the important variables in Tableau which are as under:

Histogram of Age:



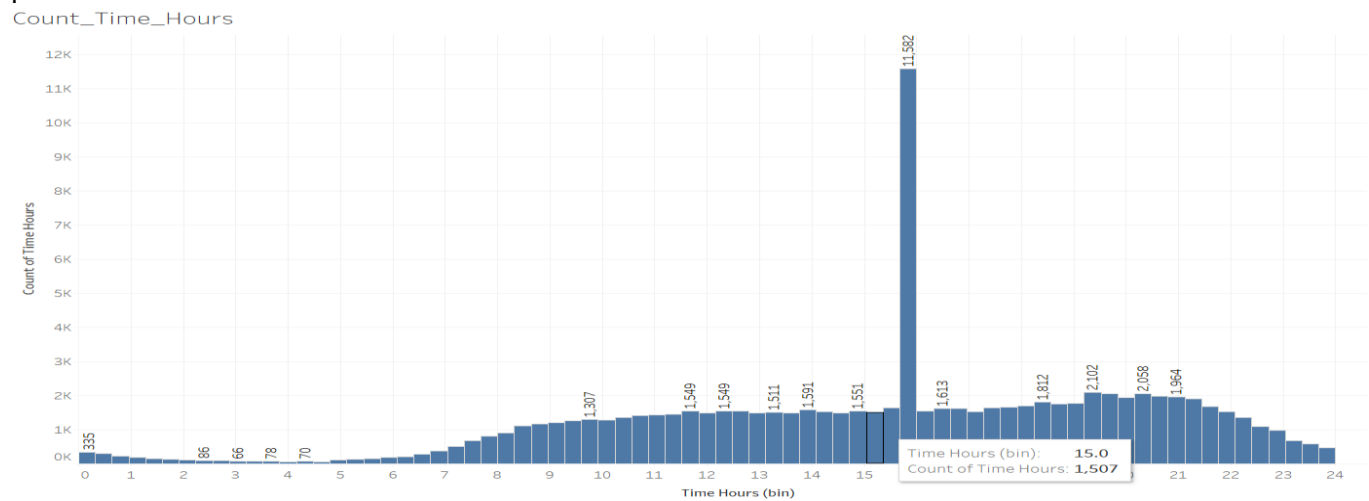
The Histogram of Age suggests that maximum customers are in the age group of 20 to 40. After the Age of 40, the count of customers reduces substantially.

Histogram of Maximum Paid Invoices:



From the Histogram of invoices, it is observed that maximum paid invoices in the last one year are in the 10K range

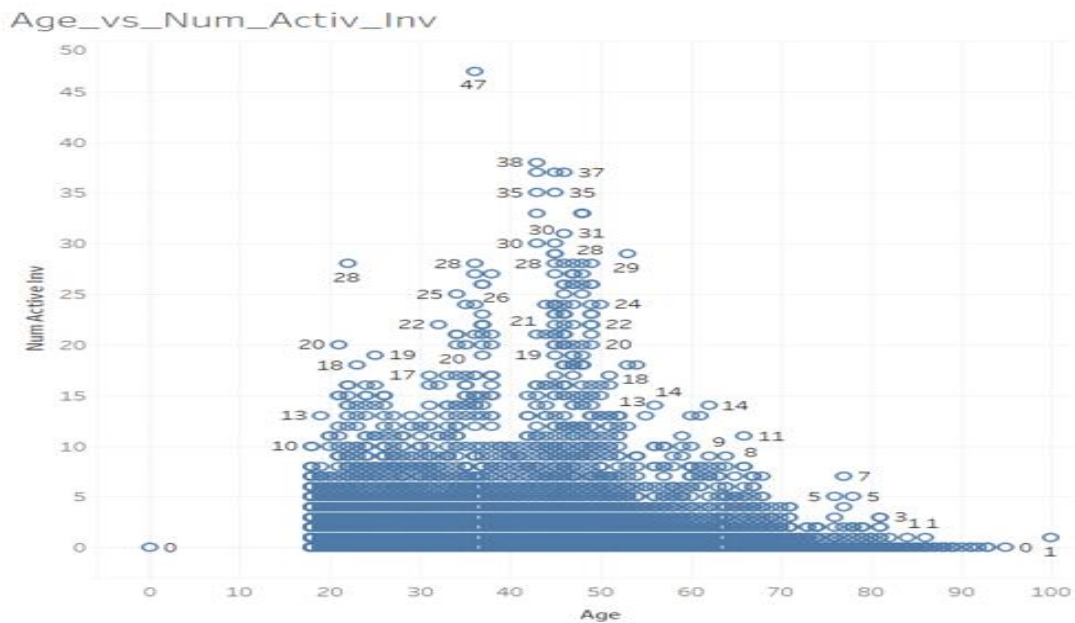
Histogram of Time in Hours:



From the Histogram of Time Hours, it is observed that maximum no of hours spent by customers doing Credit Card purchases is 16.

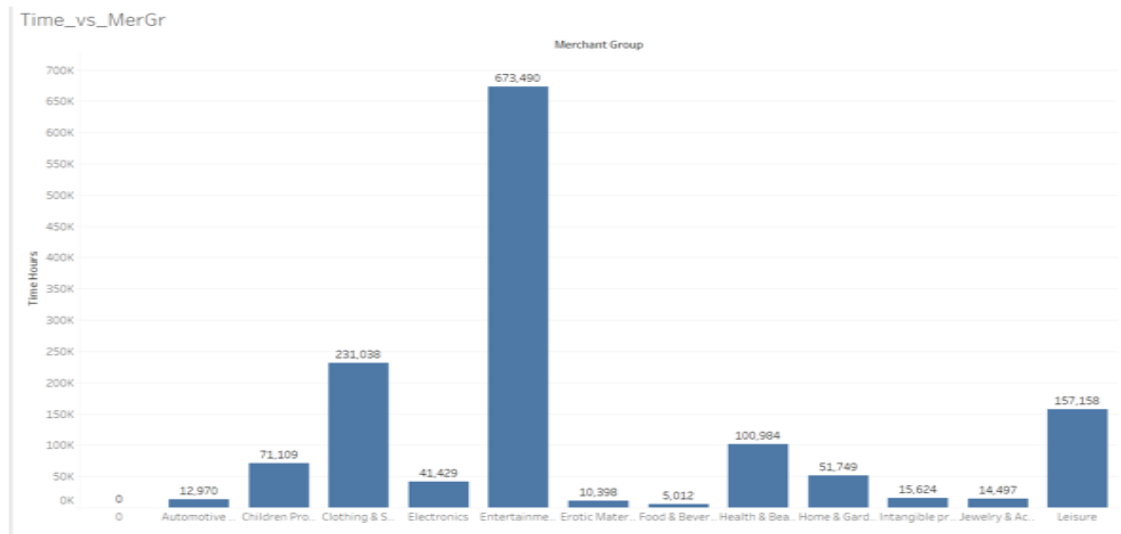
e) Bivariate Analysis:

Age vs Number of Active Invoices:



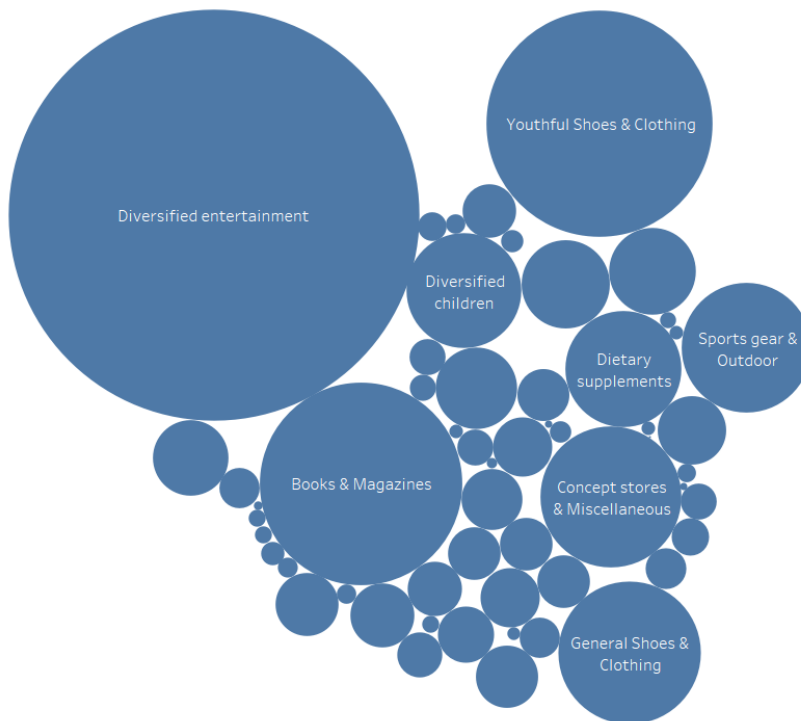
From the Scatter Plot of Age vs Number of active Invoices, it is seen that maximum number of active invoices are in the range of 40-50 age bracket.

Time vs Merchant Group:



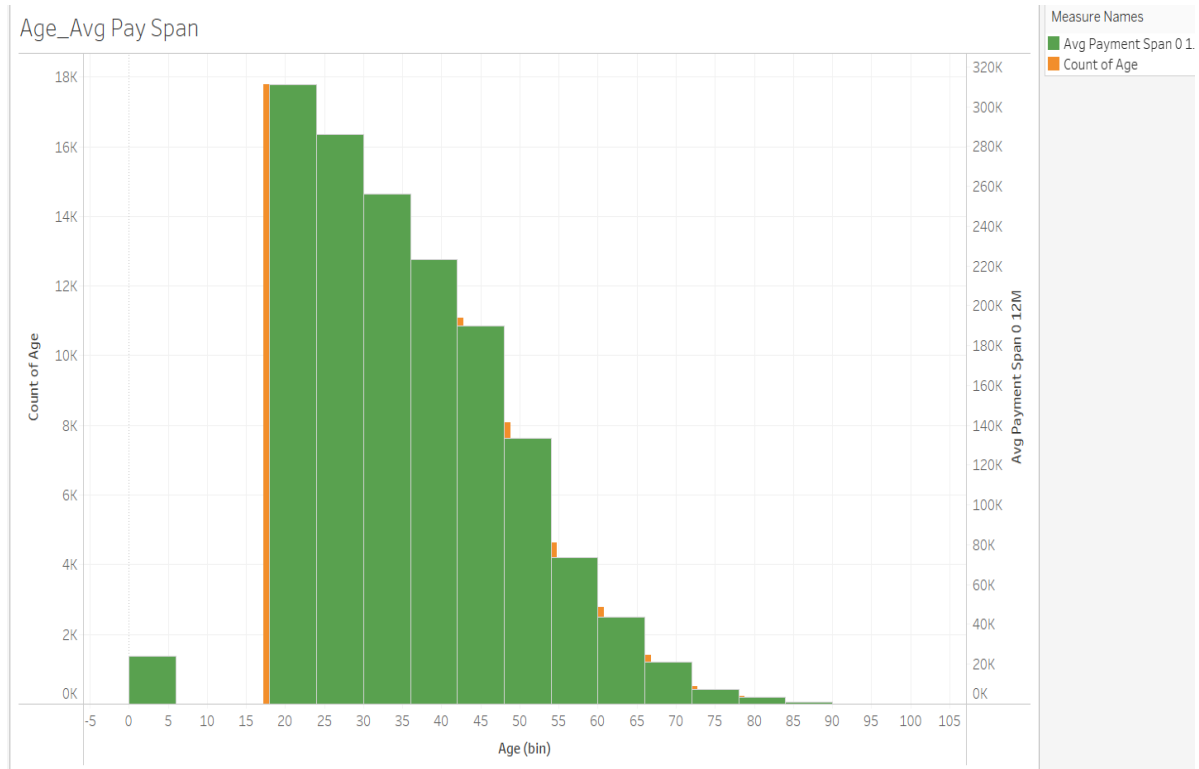
From the Bar Chart of Time in Hours vs Merchant Group, it is seen that maximum time is spent on purchase of entertainment goods.

Time vs Merchant Category:



From the Bubble Chart of Time in Hours vs Merchant Category, it is seen that maximum time is spent on purchase of diversified entertainment goods.

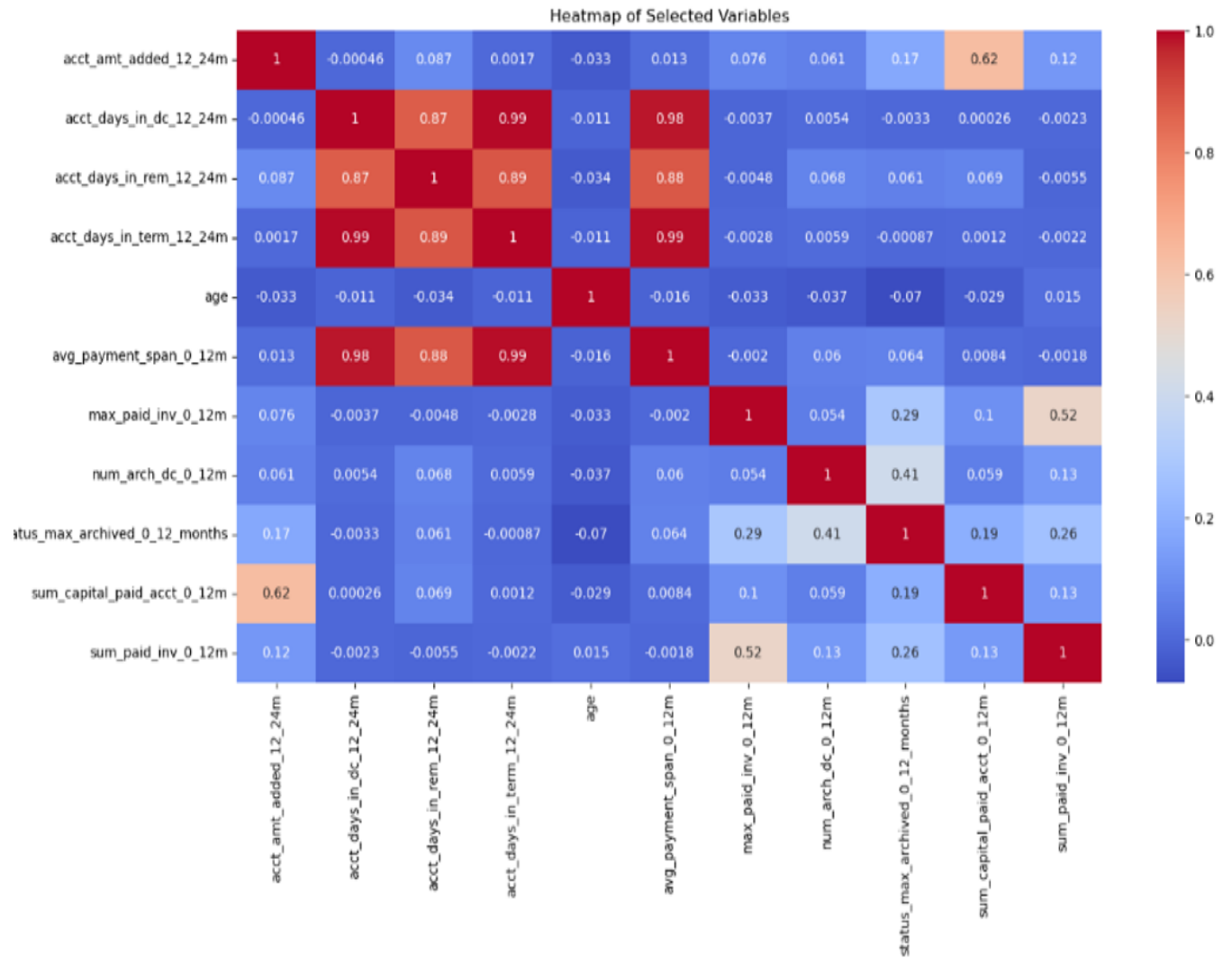
Age vs Average Payment Span in last One Year:



From the Bar Chart of Age vs Average Payment Span in last One Year, it is seen that maximum Average Payment Span in last One Year is in the Age Group of 20-25 years.

f) Multivariate Analysis:

Heatmap of Selected Variables:



Heatmap of Selected Variables was drawn as above. It is seen that only some of the variables have high multi collinearity.

3. Data Cleaning and Pre Processing:

- a) **Removal of unwanted variable:** Name in Email is considered to be an unwanted variable and is not likely to affect our analysis or add any value to analysis has been removed for more clarity.
- b) **Duplicate Values Treatment:** The data was checked for duplicate values and only one value was found duplicate. The same was removed.
- c) **Missing Values Treatment:** It has been observed that there are significant Null/Missing values in the data. The missing values in the initial data are as under:

```
acct_worst_status_12_24m      66762
acct_worst_status_6_12m       60351
acct_incoming_debt_vs_paid_0_24m  59316
acct_worst_status_3_6m        57703
acct_worst_status_0_3m        54374
acct_status                   54374
avg_payment_span_0_3m         49306
num_active_div_by_paid_inv_0_12m  29926
avg_payment_span_0_12m        23837
acct_days_in_dc_12_24m        11837
acct_days_in_rem_12_24m       11837
acct_days_in_term_12_24m      11837
num_arch_rem_0_12m            11035
max_paid_inv_0_12m            11035
status_max_archived_0_12_months  11035
num_arch_ok_12_24m            11035
status_max_archived_0_6_months  11035
sum_capital_paid_acct_12_24m    11035
status_max_archived_0_24_months  11035
recovery_debt                 11035
sum_capital_paid_acct_0_12m     11035
num_arch_dc_12_24m            11035
sum_paid_inv_0_12m            11035
num_arch_ok_0_12m             11035
time_hours                    11035
num_arch_dc_0_12m             11035
num_active_inv                11035
name_in_email                 11035
max_paid_inv_0_24m            11035
has_paid                      11035
default                       10001
merchant_group                 10
merchant_category              1
age                            1
acct_amt_added_12_24m         1
userid                         1
dtype: int64
```

Firstly, Null values of dependent variable i.e. 'default' have been dropped. From the above, it has been observed that following seven variables have more than 30% values as Null values:

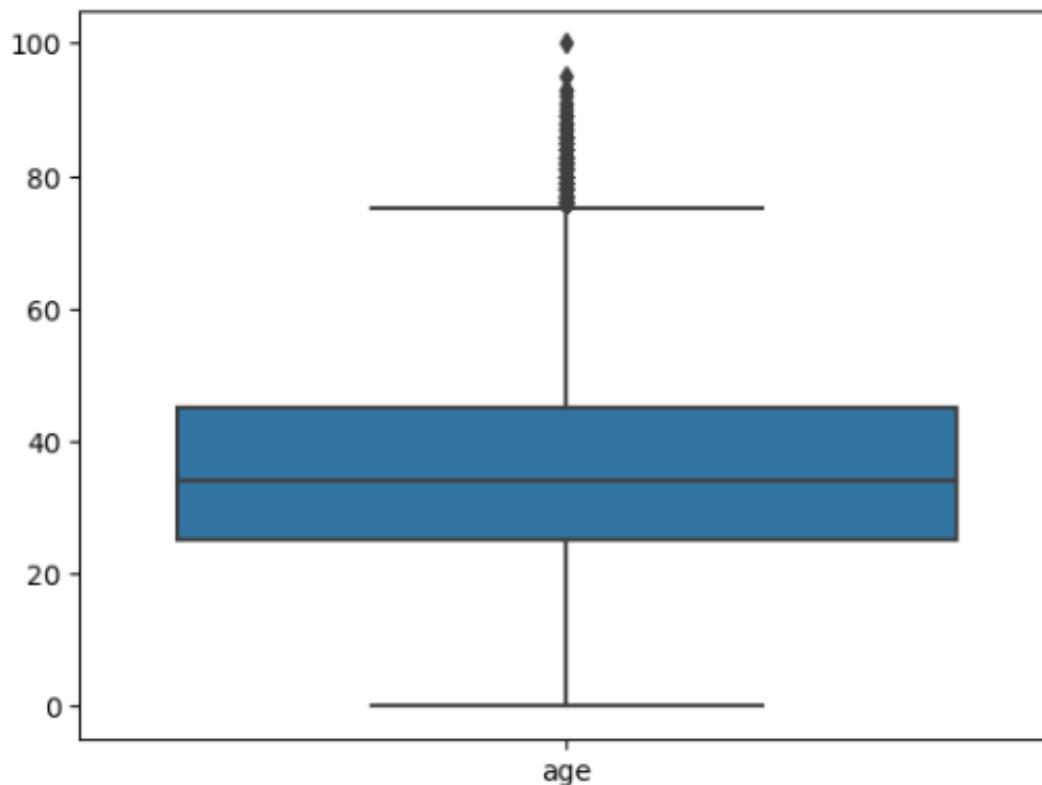
acct_worst_status_12_24m	66762
acct_worst_status_6_12m	60351
acct_incoming_debt_vs_paid_0_24m	59316
acct_worst_status_3_6m	57703
acct_worst_status_0_3m	54374
acct_status	54374
avg_payment_span_0_3m	49306

The above variables having more than 30% values as Null values have been dropped.

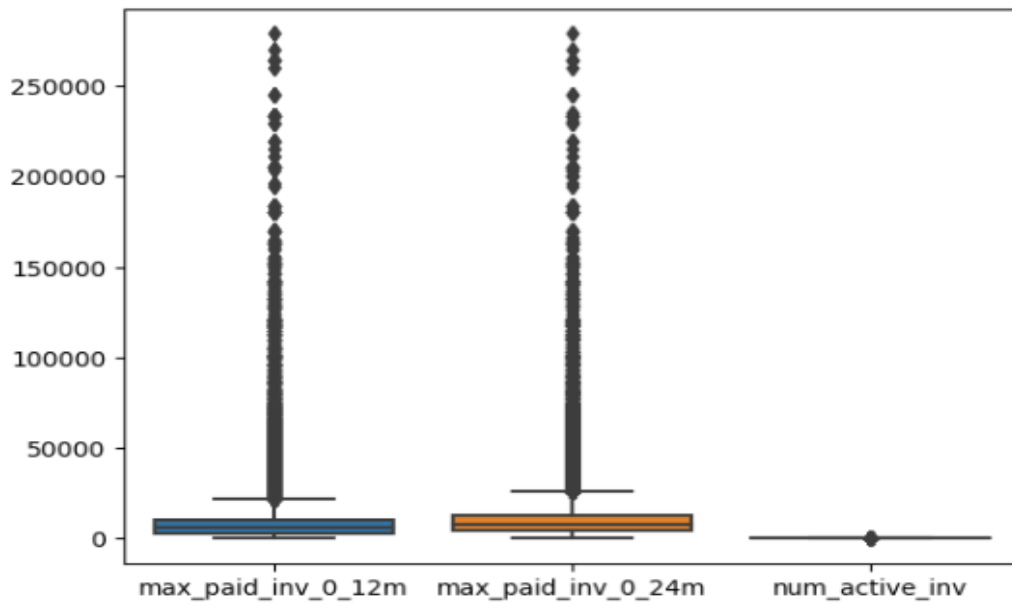
Thereafter, Null values in the continuous variables have been imputed with Median values. Further, Null values in the object type variables have been imputed with Mode values. This has been done so that important data points are not lost.

d) Outlier Treatment: Box plots of all outliers of Continuous variables have been plotted and it has been seen that there are significant outliers in the variables.

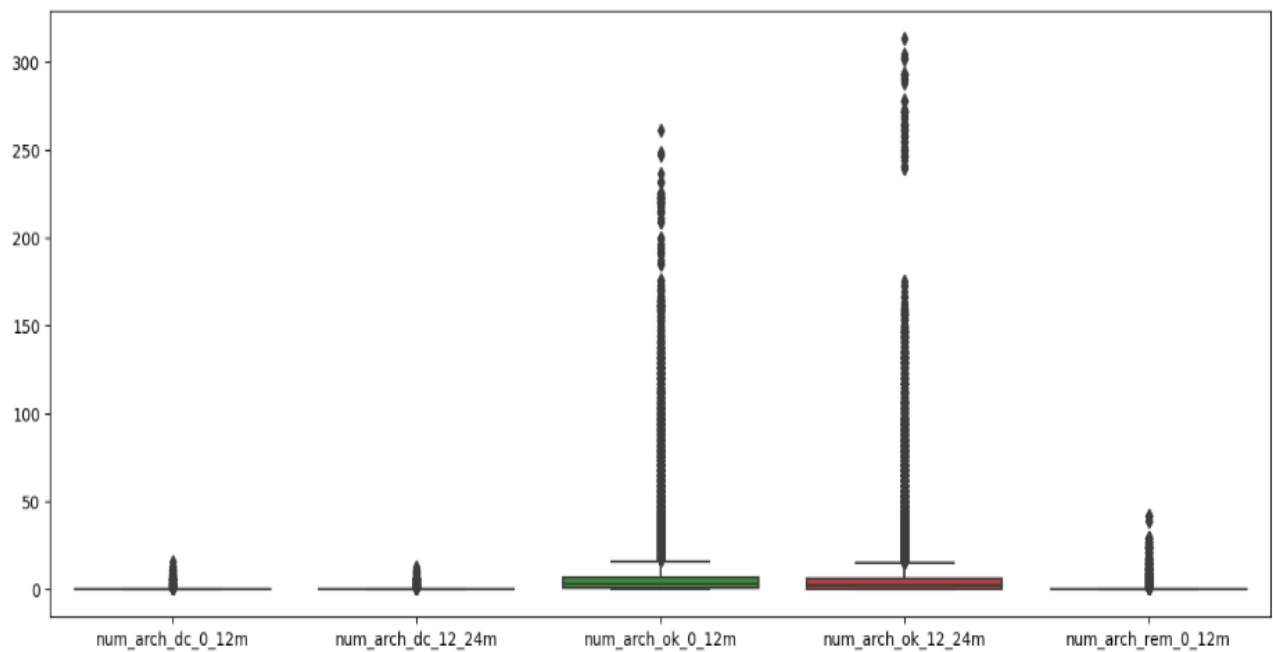
Boxplot of Age



Boxplot of Max Paid Inv 12 M, 24 M & No of Active Invoices



Boxplot of No of Archived inv in debt collection status , Ok & Reminder status in 12 M, 24 M



It has been observed that there are significant number of outliers in the data. The outliers in continuous variables have been imputed by their Median values by IQR Method so as not to lose important data points. Outliers for Object type variables are, however, not been treated.

- e) **Variable Transformation:** The object type variables such as 'default' and 'has_paid' have been transformed from float to object type.
- f) **Addition of new Variables:** Two object type variables i.e. merchant_group and merchant_category have been encoded and new variables i.e. merchant_group_encoded and merchant_category_encoded have been added and further analysis has been done with encoded variables.

4. Model Building:

After handling Null values, Outliers and EDA and cleaning of data, variables having high multicollinearity were removed by VIF Method. The variables having VIF greater than 2 were considered highly multi collinear and thus removed from model building. After this, the target variable i.e, 'default' was encoded as 0 (no default) and 1 (default).

Thereafter, Random under Sampler was applied to run the Models properly to under sample the majority class since the data is highly unbalanced.

After this, the data was split into train and test data in the ratio 70:30 and the Models were built.

The Models have been built as under:

Summary of Models:

The Summary of results of the Models with respect to default (1) parameter are as under:

Name & Type of Model	Train Data				Test Data			
	AUC Score	Precision	Recall	F1 Score	AUC Score	Precision	Recall	F1 Score
Logistic Regression (Optimized)	0.831	0.751	0.754	0.752	0.801	0.740	0.684	0.711
LDA (Optimized)	0.748	0.720	0.845	0.778	0.721	0.711	0.816	0.760
Decision Tree (Optimized)	1.00	1.00	1.00	1.00	0.671	0.692	0.641	0.666
Decision Tree (Pruned-Max Depth=3)	0.809	0.761	0.693	0.726	0.773	0.766	0.636	0.695
Bagging Classifier (Hyper Parameters Tuned with Best Parameters)	0.85	0.72	0.80	0.75	0.80	0.70	0.77	0.73
Random Forest Classifier (Hyper Parameters Tuned with Best Parameters)	0.907	0.797	0.842	0.819	0.830	0.754	0.753	0.753
ADA Boost Classifier (Hyper Parameters Tuned with Best Parameters)	0.865	0.76	0.81	0.78	0.824	0.75	0.76	0.75
Gradient Boost Classifier (Optimized)	0.930	0.828	0.875	0.851	0.833	0.756	0.745	0.751
Gradient Boost Classifier (Hyper Parameters Tuned with Best Parameters)	0.991	0.93	0.96	0.95	0.844	0.77	0.75	0.76

Random Forest Classifier (with Best Parameters) is the best Model with highest AUC Score of 0.844 on Test Data and highest F1 Score of 0.76 on Test Data (Balance between Precision and Recall)

The Model has a Train AUC Score of 0.907 and a Test AUC Score of 0.830, which is best among all the Models and also not overfitting

The Model has a Train F1 Score of 0.819 and a Test F1 Score of 0.753, which is best among all the Models.

The Model was optimized with the optimum threshold of 0.4915.

Selected Random Forest Model's hyperparameters were tuned to find the combination that results in the best performance on the validation data. The hyper tuned parameters are as under:

Best Parameters are as under:
Max Depth = 7, Min Samples Leaf = 5
Min Samples Split = 30, n estimators = 50

5. Model Validation:

Data was split into Train and Test data in 70:30 ratio i.e. Holdout validation. The Model was trained on the Train dataset and validated on test dataset.

Hyperparameter Tuning: Selected Random Forest Model's hyperparameters were tuned to find the combination that results in the best performance on the validation data. The hyper tuned parameters are as under:

Best Parameters are as under:
Max Depth = 7, Min Samples Leaf = 5
Min Samples Split = 30, n estimators = 50

Precision and Recall are the two important parameters for Credit card Default type unbalanced data problem. F1 Score, which is Harmonic Mean between Precision and Recall is 0.819 for Training Data and 0.753 for Test Data for the Default (1) parameter.

It is seen that the Model is not overfitting as the Model does not perform very poorly on Test data compared to Train data.

AUC indicates the performance of a classifier across all possible thresholds. In credit card default prediction model, a higher AUC implies means better identification of default cases, which can lead to more effective risk management decisions. The Model has AUC of 0.907 on train data and 0.830 on test data, which indicates good performance of model on all thresholds and the model is not overfitting

Feature Importance was run on the selected Model i.e. Random Forest Classifier (Best Parameters). Based on the Feature Importance, the top ten Features in order of their importance are as under:

i)	Average Payment Span 0-12 months	-	0.207983
ii)	num_arch_ok_0_12 months	-	0.179017
iii)	num_arch_ok_12_24 months	-	0.093672
iv)	status_max_archived_0_24 months	-	0.091925
v)	age	-	0.078965
vi)	sum_paid_inv_0_12 months	-	0.066631
vii)	max_paid_inv_0_12 months	-	0.056364
viii)	merchant_category_encoded	-	0.049750
ix)	merchant_group_encoded	-	0.047580
x)	status_max_archived_0_6 months	-	0.043092

6. Final interpretations / recommendations:

Final Interpretations:

The Most Optimum Model selected is the Random Forest Classifier tuned with best hyperparameters. The output predictions lead to the following insights:

- i) The Model predicts 3278 of 89977 i.e. 3.64% as possible defaulters at a default threshold of 80%.
- ii) It is observed that Average Payment Span when the Credit Card Bill becomes due in the last one year is the most important parameter for prediction of default.
- iii) It is observed that Number of Archived Purchases that were paid in the last one year is the second most important parameter for prediction of default.
- iv) From the Feature Importance of our selected Model, it is observed that Number of Archived Purchases that were paid in the last two years and the maximum number of times the account was in archived status in the last two years are third and fourth important parameters for prediction of default.
- v) From the Model, it is observed that Number of Archived Purchases that were paid in the last two years and the maximum number of times the account was in archived status in the last two years are third and fourth important parameters for prediction of default.
- vi) From the Model, it is observed that Age of the Customer is also important parameters for prediction of default. It has been seen that customer of 18 to 25 Age group have maximum Average Payment Span in the last one year.

Recommendations:

Based on the above, the recommendations are as under:

- i) 3278 customers (3.64%) who are predicted as potential defaulters are to be closely monitored for any signs of possible default by the company.
- ii) Average Payment Span of the customer when the Credit Card bill becomes due in the last one year is to be tracked very closely by the Company for all customers.
- iii) Number of Archived Purchases that were paid in the last one year is to be tracked very closely by the Company for all customers.
- iv) Number of Archived Purchases that were paid in the last two years and the maximum number of times the account was in archived status in the last two years are also be tracked very closely by the Company for all customers.
- v) It has been seen that customer of 18 to 25 Age group have maximum Average Payment Span in the last one year, which is most important parameter as per Feature Importance. Hence, customers of 18 to 25 age group are to be tracked very closely by the Company.
- vi) The Credit Limits, Credit ratings and hot listings of the customers may be done for the predicted 3.64% default cases from the Model results.

The above measures will eventually lead to higher profitability and less NPAs for Credit Card Company.